MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER BPES EXAMINATION (2024 ADMISION ONWARDS)

COURSE CODE – MG3DSCPES201 COURSE TITLE UNDERSTANDING HUMAN BODY

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions Answer All Questions Each question carries 1 mark

1. Which of the following is not a basic type of human tissue? [U] (C O-1)
    a. Epithelial
    b. Connective
    c. Skeletal
    d. Nervous
2. The shoulder joint is classified as which type of synovial joint? [U] (C O-2)
    a. Hinge
    b. Pivot
    c. Ball and socket
    d. Saddle
3. Which characteristic of muscle tissue refers to its ability to return to its original length after being stretched? [U] (C O-2)
    a. Contractibility
    b. Elasticity
    c. Excitability
    d. Irritability
4. What structure connects muscles to bones? [U] (C O-1)
    a. Ligaments
    b. Cartilage
    c. Tendons
    d. Discs
5. Which anatomical term describes a movement that increases the angle between two bones? [A] (C O-3)
    a. Flexion
    b. Extension
    c. Abduction
    d. Rotation
6. The femur is classified as a: ---------- [U] (C O-1)

a. Short bone
b. Irregular bone
c. Flat bone
d. Long bone

7. Which part of the brain is responsible for balance and coordination? [U] (C O -2)
    a. Cerebrum
    b. Cerebellum
    c. Medulla oblongata
    d. Thalamus
8. The sliding filament theory explains: [U] (C O-2)
    a. How bones grow
    b. How blood circulates
    c. How neurons transmit signals
    d. How muscles contract
9. In a reflex arc, the role of the spinal cord is to: [A] (C O-4)
    a. Initiate voluntary movement
    b. Interpret visual signals
    c. Directly process and respond to stimuli
    d. Store memory
10. Why is understanding posture important for sports performance and injury prevention? [A] (C O-6)
    a. It improves heart rate
    b. It reduces muscle mass
    c. It enhances coordination and minimizes strain
    d. It increases bone density

PART – B

Short Answer any 10 questions Each questions carries 2 Marks

11. Define the term *anatomy* and mention its importance in physical education. [U] (C O-1)
12. Name the four basic types of tissues found in the human body. [U] (C O-1)
13. Write any two functions of red blood cells. [U] (C O-2)
14. Differentiate between axial and appendicular skeleton. [U] (C O-1)
15. Mention any two structural features of the *humerus*.[S] (C O-5)
16. Define a *synovial joint* and give one example. [U] (C O-2)
17. What do you mean by *flexion* and *extension*? Give one example of each movement. [A] (C O-3)
18. Write two general characteristics of skeletal muscles. [U] (C O-2)

19. Mention any two differences between slow-twitch and fast-twitch muscle fibers. [U] (C O-2)
20. Name the three types of nerves found in the human nervous system. [U] (C O-2)
21. Write two major functions of the cardiovascular system. [U] (C O-2)
22. How does anatomical knowledge help in maintaining good posture? [A] (C O-6)

PART – C

Short Essays Answer any 4 questions Each questions carries 5 marks

23. Explain the structure and functions of any two types of tissues in the human body with examples. [U] (C O-1)
24. Describe the classification of bones with suitable examples. Mention one functional aspect of each type. [U] (C O-2)
25. Discuss the structure of the synovial joint using the knee joint as an example. Include its components and types of movements allowed. [An] (C O-2)
26. Describe the sliding filament theory of muscle contraction and mention its significance in sports movement. . [An] (C O-2, 4)
27. Illustrate how the structure of the brain and spinal cord supports reflex action in the human body. . [U] (C O-2)
28. Explain the role of anatomical knowledge in maintaining correct posture and preventing musculoskeletal injuries in sports. . [A, Ap] (C O-4, 6)

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